

Bayesian Hierarchical Model for PM_{2.5} levels in Italy

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1 Introduction

Air pollution represents a major environmental and public health concern. Specifically, fine particulate matter $\text{PM}_{2.5}$ is among the most harmful pollutants due to its ability to penetrate deep into the respiratory system and its strong association with cardiovascular and pulmonary diseases.

A key challenge in modeling $\text{PM}_{2.5}$ concentrations lies in the presence of substantial heterogeneity across monitoring stations and irregular temporal coverage. Stations differ significantly in terms of geographical characteristics, local emission sources, and environmental conditions, while observational data often exhibit an unbalanced panel structure, with varying numbers of observations across stations and missing time periods.

In this context, Bayesian hierarchical models provide a natural and flexible framework to account for these complexities. By introducing group-level random effects, these models allow for partial pooling, enabling information sharing across stations while preserving local heterogeneity. Moreover, the Bayesian approach provides a coherent framework for uncertainty quantification and prior specification.

This paper develops a Bayesian hierarchical model for daily $\text{PM}_{2.5}$ concentrations observed across monitoring stations in Italy. Posterior inference is carried out using a Metropolis-within-Gibbs sampler. The analysis focuses on model specification, MCMC implementation, convergence diagnostics, sensitivity analysis, and posterior predictive checks.

2 Data

The dataset comprises daily $\text{PM}_{2.5}$ concentration measurements collected across monitoring stations in Italy during 2025. The raw data were sourced from the European Environment Agency (EEA) Air Quality portal, specifically via the EEA Air Quality e-Reporting, with metadata and station specifications validated through the EEA Air Quality Viewer. After preprocessing and quality control, the final dataset contains 31,349 observations distributed across 116 stations.

The empirical distribution of the raw outcome variable is highly right-skewed, with a skewness of approximately 17.1. This strong asymmetry is primarily driven by extreme pollution events in the upper tail. To stabilize the variance and satisfy the distributional assumptions of the proposed model, a logarithmic transformation was applied

$$\log(\text{PM}_{2.5,it}) = y_{it}$$

where y_{it} represents the transformed concentration for station i at time t . This transformation substantially reduces skewness (to approximately -0.29), yielding a more symmetric distribution (visible in Figure 1) that justifies the use of a Gaussian likelihood.

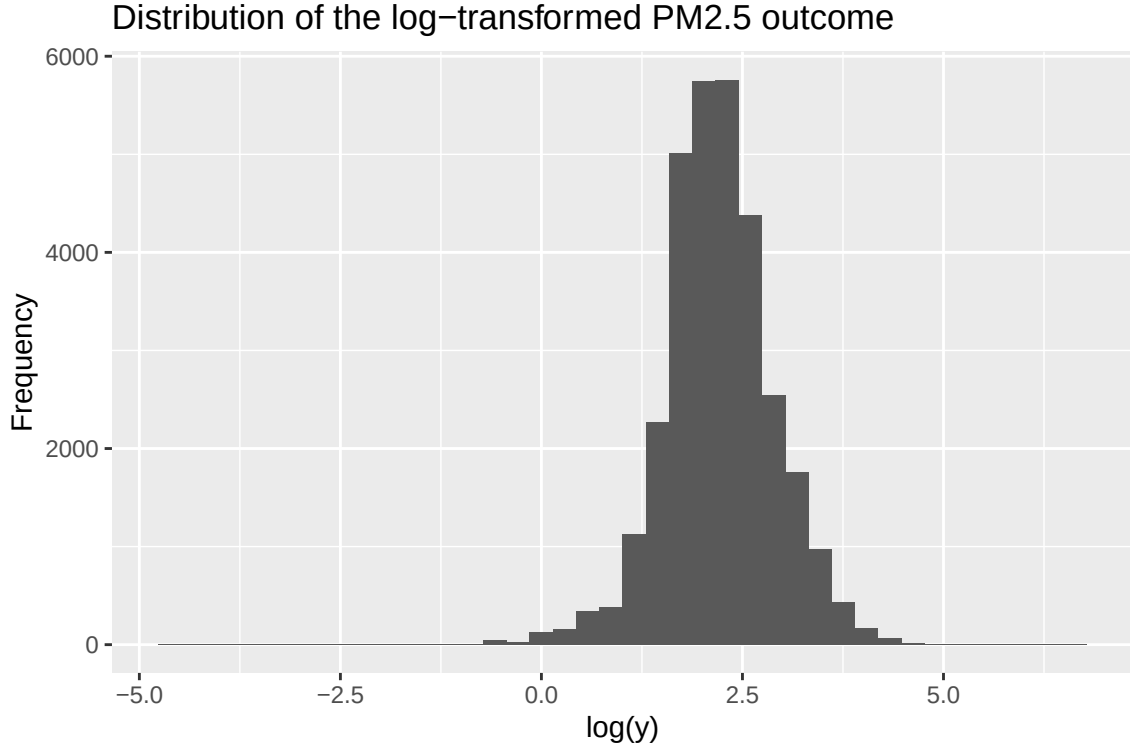


Figure 1: Distribution of the log-transformed PM2.5 concentrations.

The panel structure is markedly unbalanced, a common feature in environmental monitoring networks. The number of observations per station ranges from 69 to 332, with a median of approximately 299. Temporal coverage is also uneven across stations. While most stations provide observations for most of the year, 32 stations lack at least one full month of data, and for a few stations coverage drops to only 4 months. This heterogeneity is summarized in Figure 2, which reports both the distribution of the number of observations per station and the distribution of the number of months observed per station. Such an irregular structure poses a challenge for standard time-series methods.

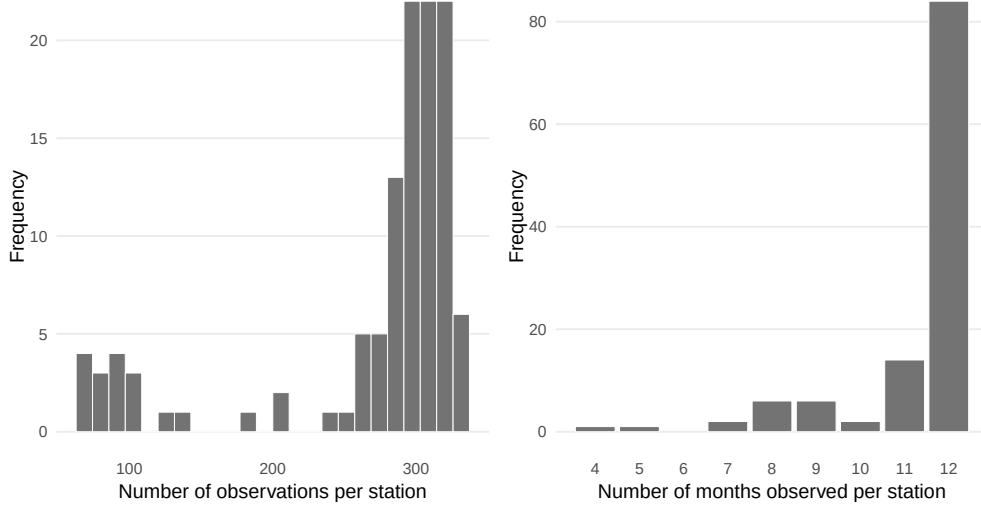


Figure 2: Data availability by station and month. Darker areas indicate months with fewer observations.

3 Model

The hierarchical nature of the data, together with the strong imbalance in the panel structure, naturally motivates a multilevel modeling approach. In particular, the uneven number of observations across stations and the presence of incomplete temporal coverage make standard time-series or pooled regression models potentially inefficient and biased, as they are unable to properly account for unobserved heterogeneity across monitoring sites.

To address these issues, a Bayesian hierarchical specification is adopted, which allows for partial pooling across stations. This framework enables the model to borrow strength from the entire dataset while retaining station-specific components, thus providing robust inference even in the presence of sparse observations for some units.

A hierarchical model is considered for the log-transformed $\text{PM}_{2.5}$ concentrations. Let $\tilde{y}_i$ denote the transformed outcome for observation i . The likelihood is specified as:

$$y_{it} \mid \beta, \alpha, \sigma^2 \sim \mathcal{N}(x_{it}^\top \beta + \alpha_i, \sigma^2)$$

where:

- x_i is a vector of covariates including month and day-of-week effects,
- β is a vector of regression coefficients capturing systematic temporal variation,
- $\alpha_{j[i]}$ is a station-specific random intercept accounting for time-invariant local characteristics.

3.1 Hierarchical structure

Station-specific effects are modeled as draws from a common Gaussian population distribution:

$$\alpha_j \mid \tau^2 \sim \mathcal{N}(0, \tau^2), \quad j = 1, \dots, J,$$

where τ^2 captures the cross-station variability in baseline pollution levels.

The assumption of a common Gaussian distribution reflects the idea that monitoring stations are exchangeable draws from a broader population of locations, once observed covariates are accounted for. This formulation allows the model to decompose variation in pollution levels into a systematic component (captured by $x_i^\top \beta$) and an unobserved station-specific component.

The hierarchical structure induces partial pooling: stations with limited observations are shrunk toward the overall mean, while stations with richer data retain more idiosyncratic estimates. This feature is particularly important in the present setting, where the panel is highly unbalanced and some stations provide only sparse temporal coverage. Shrinkage improves estimation stability and prevents overfitting to noisy local patterns.

3.2 Prior specification

Weakly informative priors are assigned to all parameters to regularize estimation while allowing the data to drive inference.

For the regression coefficients:

$$\beta \sim \mathcal{N}(b_0, B_0),$$

where $B_0 = cI$ is a diagonal covariance matrix. This specification reflects prior independence across coefficients and imposes mild shrinkage toward zero, helping to stabilize estimation in the presence of potentially correlated temporal indicators.

The observation variance is assigned the prior

$$\sigma^2 \sim \text{Inverse-Gamma}(a_\sigma, b_\sigma).$$

which provides a conjugate prior facilitating posterior computation while remaining weakly informative for suitably chosen hyperparameters.

For the group-level standard deviation:

$$\tau \sim \text{Half-Cauchy}(0, A).$$

The use of a half-Cauchy prior for τ avoids the well-known sensitivity of inverse-gamma priors for variance components, while providing weakly informative regularization.

3.3 Posterior inference

Posterior inference is carried out via a Metropolis-within-Gibbs sampler. Conditional on the variance components, the regression coefficients β and the random effects α admit standard Gaussian full conditional distributions and are updated via Gibbs steps. The observation variance σ^2 is sampled from its inverse-gamma full conditional distribution.

The group-level scale parameter τ , for which no closed-form full conditional is available under the half-Cauchy prior, is updated using a random-walk Metropolis step. This hybrid scheme combines computational efficiency for conjugate blocks with flexibility in handling non-conjugate components.

The sampling scheme can be summarized as follows:

- Sample β from its conditional posterior
- Sample each α_j from its conditional posterior
- Sample σ^2 from its conditional posterior
- Update τ via a Metropolis-Hastings step

3.4 Implementation details

Hyperparameters are selected to induce weakly informative priors. For the regression coefficients, the prior covariance matrix is set to $B_0 = cI$ with a large value of c . For the observation variance, the inverse-gamma hyperparameters a_σ and b_σ are chosen to yield a diffuse prior. The scale parameter of the half-Cauchy prior on τ is set to A , allowing for substantial between-station variability while still providing regularization.

Posterior inference is performed using a Metropolis-within-Gibbs algorithm. The sampler is run for 5,000 iterations, with the first 200 discarded as burn-in. To reduce autocorrelation, the chain is thinned by retaining every 5th draw.

The Metropolis step for τ is implemented as a Gaussian random walk on the log scale. A tuning exercise was conducted to calibrate the proposal variance. With a proposal standard deviation of 0.15, the acceptance rate was approximately 0.48, while increasing it to 0.20 reduced the rate to approximately 0.38. Since posterior summaries and diagnostic plots were essentially unaffected, the larger proposal scale is retained to improve exploration of the parameter space.

Convergence is assessed via trace plots and autocorrelation functions, which indicate satisfactory mixing across all parameters. Effective sample sizes are sufficiently large across parameter blocks, with values around 900 for the regression coefficients and above 700 for the variance components. For the station-specific effects, effective sample sizes are generally above 70, reflecting slower mixing but still adequate precision given the dimensionality of the parameter space. Overall, the sampler exhibits satisfactory mixing and efficiency.

4 Results

4.1 Convergence diagnostics

Trace plots of the post-burn-in chains (Figure 3) show no evidence of non-stationarity or systematic trends. The chains fluctuate around stable levels, indicating satisfactory convergence of the Metropolis-within-Gibbs sampler.

Residual autocorrelation is illustrated in Figure 4. While mild dependence persists for some parameters, it is sufficiently limited and is further mitigated through thinning.

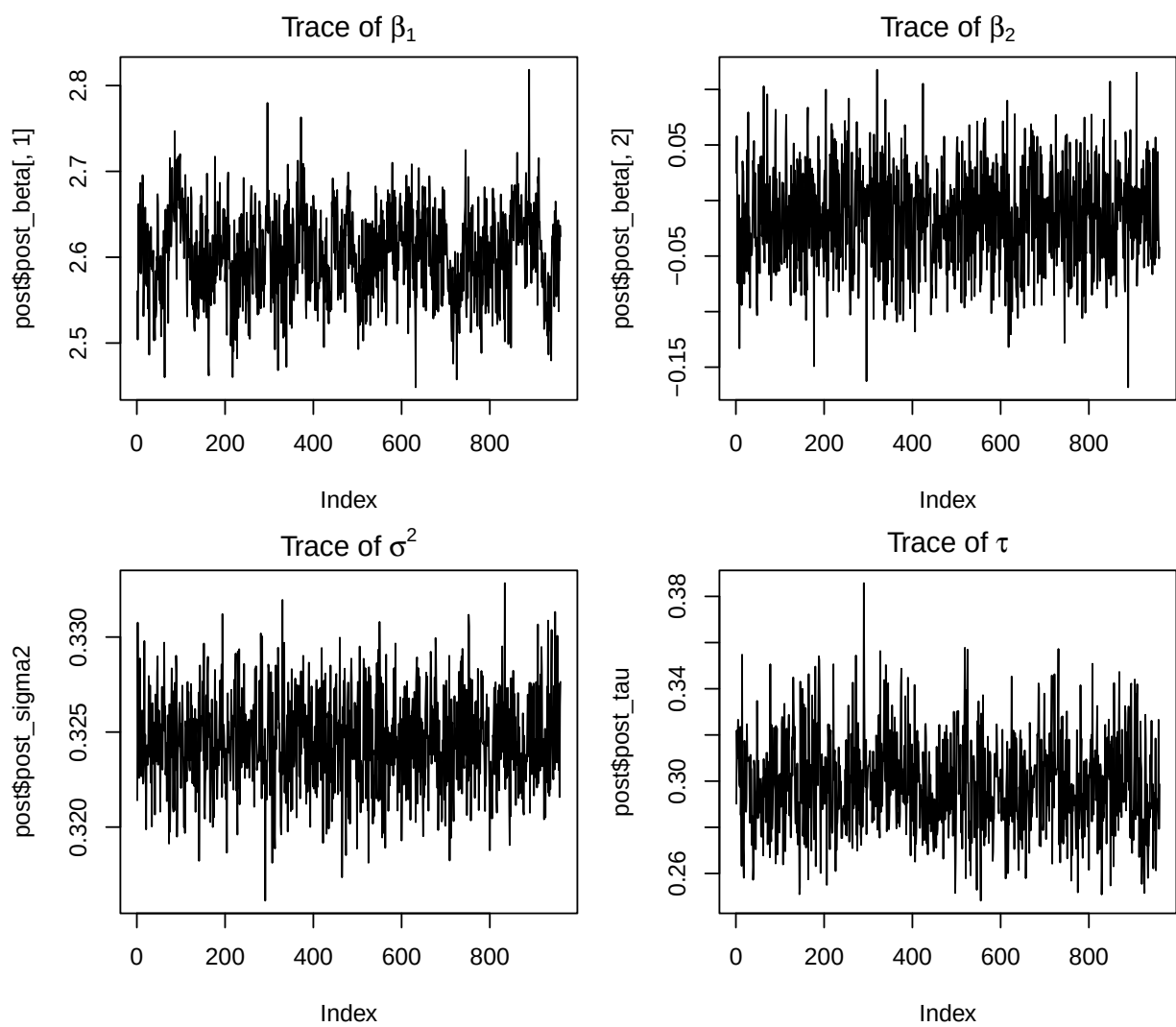


Figure 3: Post-burn-in trace plots for selected model parameters.

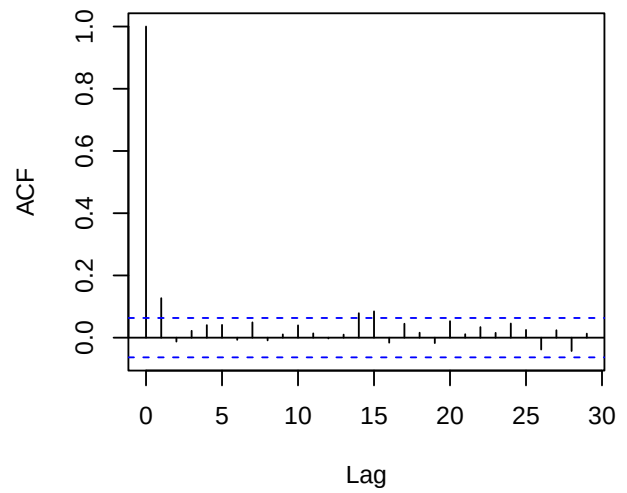
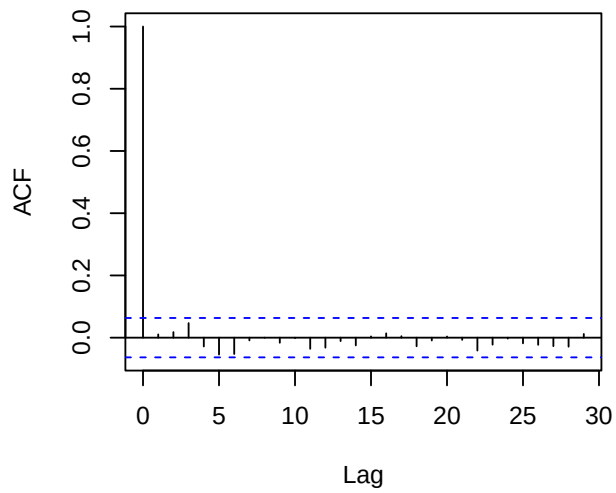
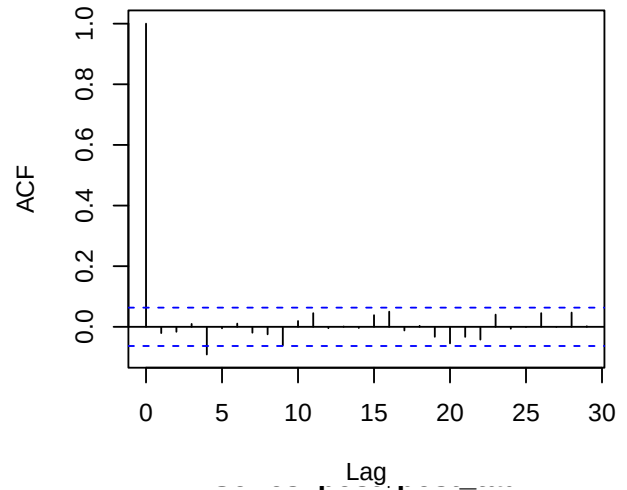
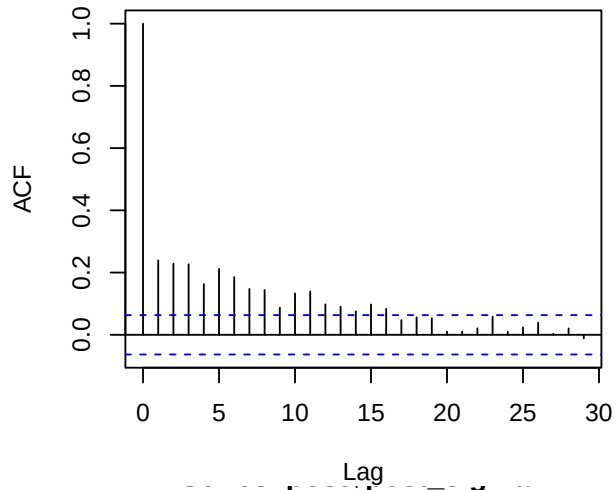


Figure 4: Post-burn-in autocorrelation functions for selected model parameters.

The Geweke diagnostic confirms these findings, with all z-scores well within conventional acceptance ranges.

Effective sample sizes are generally high across parameter blocks. In particular, regression coefficients exhibit values close to the total number of retained draws, while the variance components also show strong efficiency. Station-specific effects display lower ESS, as expected given their dimensionality, but remain within acceptable ranges. Overall, the sampler exhibits satisfactory mixing and efficiency.

4.2 Posterior summaries

The posterior estimates shown in Table 1 provide a clear and economically meaningful interpretation of the main drivers of PM2.5 concentrations.

Table 1 reports posterior means for the main coefficients.

The intercept represents the expected value of the log-transformed outcome for the reference category (January, Monday) at an average station. Its posterior mean is approximately 2.60, corresponding to a baseline PM2.5 level of about 13.5 on the original scale.

4.2.1 Seasonal effects

A strong seasonal pattern emerges. Relative to January, nearly all months exhibit negative coefficients, indicating substantially lower pollution levels throughout most of the year.

The magnitude of these effects is large: spring, summer, and early autumn months are associated with reductions often exceeding 30–50%. December appears statistically close to January, suggesting a return to high pollution levels in winter.

This pattern is consistent with known environmental mechanisms, including increased heating emissions and reduced atmospheric dispersion during colder months.

In contrast, day-of-week effects are negligible: coefficients are small and their credible intervals include zero, indicating no systematic weekly cycle once seasonal and station-level heterogeneity are accounted for.

4.2.2 Variance decomposition

The posterior estimate of the residual standard deviation exceeds the group-level standard deviation, indicating that within-station variability dominates over persistent cross-station differences. Nonetheless, the latter remains non-negligible, confirming the presence of structural heterogeneity.

Table 1: Posterior means and approximate percentage effects relative to the baseline (January, Monday).

Variable	Mean	Effect
Intercept (Jan, Mon)	2.603	Baseline (~13.5)
February	-0.016	-1.6%
March	-0.362	-30.4%
April	-0.615	-45.9%
May	-0.742	-52.4%
June	-0.119	-11.3%
July	-0.598	-45.0%
August	-0.425	-34.6%
September	-0.799	-55.0%
October	-0.605	-45.4%
November	-0.381	-31.7%
December	0.068	+7.0%

4.2.3 Spatial analysis

Figure 5 displays the geographical distribution of monitoring stations, with colors representing the posterior mean of the station-specific random effects.

The analysis is exploratory and based on a simple latitude-based grouping rather than a formal spatial model. Positive values indicate structurally higher PM2.5 levels after controlling for temporal effects.

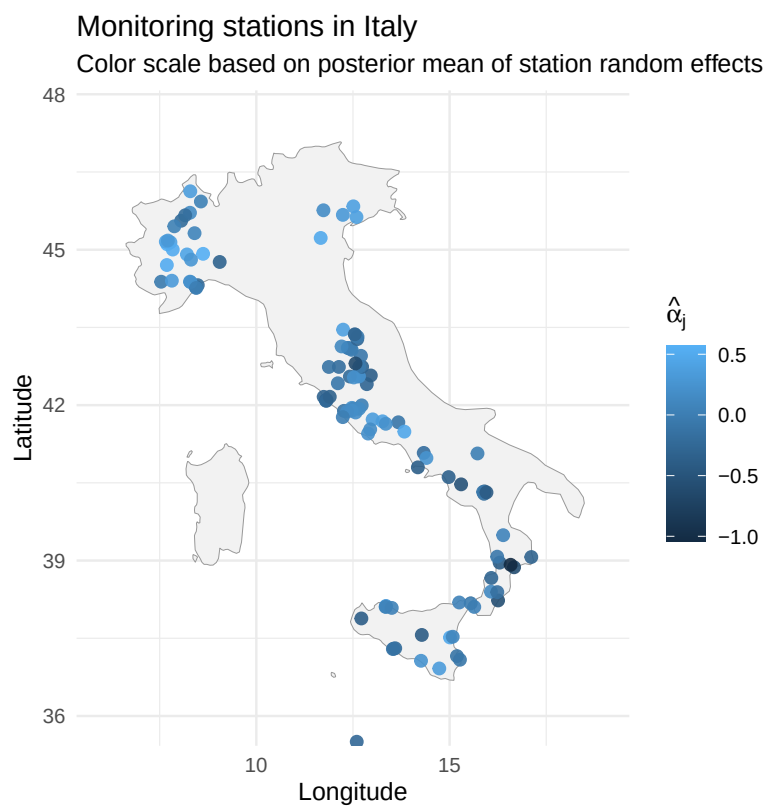


Figure 5: Spatial distribution of posterior mean station effects across Italy.

The spatial distribution reveals a clear North–South gradient in the station-level effects. Stations located in northern Italy tend to exhibit positive random effects, whereas southern stations display predominantly negative values, with central regions lying in between.

This pattern is confirmed by a posterior comparison of average station effects across macro-areas. The North–South difference is estimated at approximately 0.31 on the log scale, with a 95% credible interval entirely above zero. The posterior probability that the North exhibits higher baseline pollution than the South is effectively equal to one.

On the original scale, this corresponds to roughly a 36% higher structural level of $\text{PM}_{2.5}$ in the North relative to the South.

This evidence is consistent with well-documented geographical differences in pollution dynamics, including higher industrial intensity, traffic exposure, and less favorable atmospheric dispersion conditions in northern regions. At the same time, the dispersion of the station effects highlights substantial within-region heterogeneity, suggesting that local factors remain important determinants of air quality.

4.3 Posterior predictive checks

Posterior predictive checks were performed to assess model adequacy. Replicated datasets were generated from the posterior distribution and compared to the observed data.

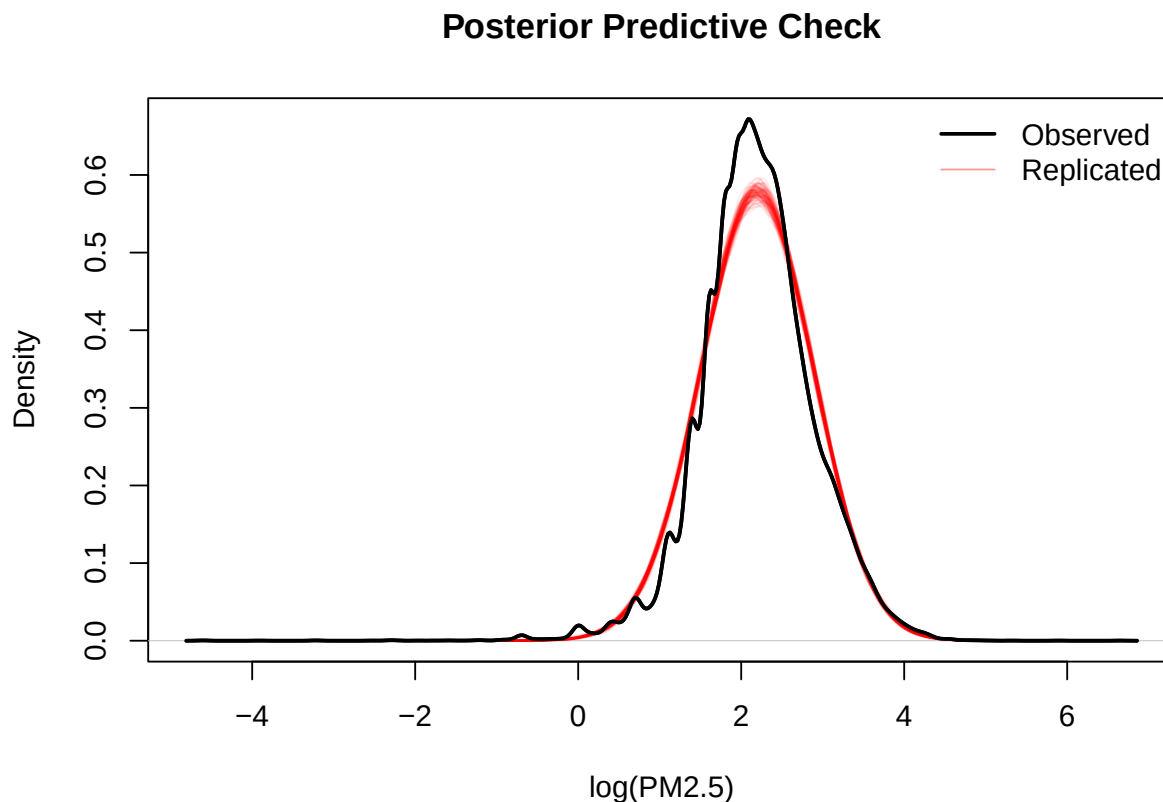


Figure 6: Posterior predictive check for the log-transformed $\text{PM}_{2.5}$ outcome.

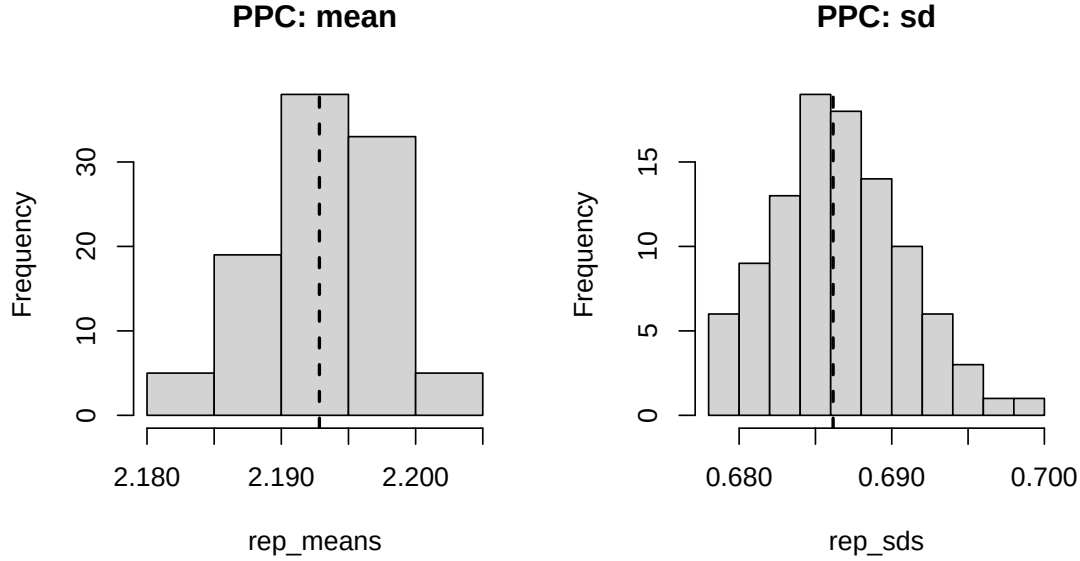


Figure 7: Posterior predictive checks based on summary statistics. The observed mean and standard deviation fall well within the distribution of replicated values.

The distribution of replicated means and standard deviations is consistent with the observed values (see Figures @ref(fig:fig-ppc-density and 7, indicating that the model accurately captures the first two moments of the data.

Furthermore, the posterior predictive density aligns well with the empirical distribution of log-transformed PM2.5, with only minor discrepancies in the peak and tail regions. These differences are consistent with the Gaussian assumption and do not indicate major model misspecification.

4.4 Sensitivity analysis

A one-at-a-time sensitivity analysis was conducted to assess the robustness of posterior inference to alternative prior specifications. Starting from the baseline configuration $c_{\text{prior}} = 100$, $A_{\tau} = 5$, and $b_{\sigma} = 0.5$, each hyperparameter was varied separately while keeping the others fixed.

In particular, three dimensions of prior calibration were examined. First, the prior variance of the regression coefficients was modified by considering $c_{\text{prior}} = 10$ and $c_{\text{prior}} = 1000$, in place of the baseline value 100. Second, the scale parameter of the half-Cauchy prior on the group-level standard deviation was varied from $A_{\tau} = 5$ to $A_{\tau} = 2.5$ and $A_{\tau} = 10$. Third, the scale parameter of the inverse-gamma prior on the observation variance was increased from $b_{\sigma} = 0.5$ to $b_{\sigma} = 1$, while keeping $a_{\sigma} = 2$ fixed.

The results are stable across all specifications. In particular, the posterior mean of the intercept, as well as the posterior means of σ^2 and τ , remain very close to their baseline values under all alternative priors considered. No meaningful changes emerge in the substantive interpretation of the temporal effects or in the magnitude of the station-level heterogeneity.

This evidence suggests that posterior inference is not materially driven by the specific prior calibration adopted in the baseline model. Rather, the likelihood appears to dominate the posterior

Table 2: Sensitivity analysis under alternative prior specifications.

Scenario	c_prior	A_tau	b_sigma	Intercept	Sigma ²	Tau	Acceptance (tau)
Baseline	100	5.0	0.5	2.603	0.325	0.299	0.388
B0 smaller	10	5.0	0.5	2.602	0.325	0.298	0.394
B0 larger	1000	5.0	0.5	2.603	0.325	0.299	0.401
A_tau smaller	100	2.5	0.5	2.603	0.325	0.299	0.384
A_tau larger	100	10.0	0.5	2.603	0.325	0.299	0.388
b_sigma larger	100	5.0	1.0	2.603	0.325	0.298	0.393

distribution, indicating that the data are sufficiently informative for the parameters of interest.

From a computational perspective, acceptance rates for the Metropolis step on τ also remain within a satisfactory range across specifications, indicating that the sampler retains stable performance under moderate prior perturbations.

Table 2 reports the posterior means of the main parameters under the alternative prior specifications considered.

5 Conclusion

This paper develops a Bayesian hierarchical model for daily PM2.5 concentrations, accounting for heterogeneity across monitoring stations and unbalanced panel structure.

The Metropolis-within-Gibbs sampler exhibits good convergence properties, supported by multiple diagnostic tools. Posterior inference is robust to prior specification, and posterior predictive checks indicate a satisfactory model fit.

The results highlight a strong seasonal component in PM2.5 concentrations, with substantially higher levels during winter months. This supports the relevance of seasonally targeted environmental policies, particularly those addressing emissions from heating and transport.

At the same time, the absence of a meaningful weekly cycle suggests that policies based solely on short-term temporal restrictions may have limited effectiveness unless combined with broader structural interventions.

Finally, the presence of persistent differences across monitoring stations, as well as the identified North–South gradient, underscores the importance of geographically differentiated policy approaches.

Overall, the hierarchical Bayesian framework provides a flexible and effective approach for modeling air pollution data. Possible extensions include the incorporation of spatial dependence across stations, the use of more flexible likelihood specifications, and the inclusion of additional covariates capturing environmental and meteorological factors.